Daily Literature Review — 2026-09-08

Daily Literature Review: AI and machine learning in protein function prediction

Date: 2026-09-08

Research Question

How are machine-learning methods being used to predict protein function, what datasets and evaluation metrics are used, and what limitations remain?

Summary

This document compiles 40 recent papers related to the topic. AI drafting was unavailable, so the content below is built directly from bibliographic metadata. Each entry includes title, authors, year, source, DOI/URL, and a reminder to extract findings from the original paper. This is an evidence-collection scaffold, not a finished scientific review.

Evidence Table (Bibliographic)

2026

Rapid directed evolution guided by protein language models and epistatic interactions.

- Authors: Tran VQ, Nemeth M, Bartie LJ, Chandrasekaran SS, Fanton A, Moon HC, Hie BL, Konermann S

- Source: PubMed

- PMID: 41712694

- Link: https://pubmed.ncbi.nlm.nih.gov/41712694/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Discovery of natural RORγt inhibitor using machine learning, virtual screening, and in vivo validation.

- Authors: Yoo H, Han SJ, Lee JE, Cho C, Hong D, Jeong B, Kim S, Jung GY

- Source: PubMed

- PMID: 40915561

- Link: https://pubmed.ncbi.nlm.nih.gov/40915561/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2025

LCN2 drives ferroptosis-associated ischemia-reperfusion injury after renal transplantation: integrated machine learning and in vivo validation.

- Authors: Wu Z, Yu B, He Q, Huang C

- Source: PubMed

- PMID: 41205031

- Link: https://pubmed.ncbi.nlm.nih.gov/41205031/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Deciphering the mechanism of baicalein in cervical cancer via bioinformatics, machine learning and computational simulations: PIM1 and CDK2 are key target proteins.

- Authors: Wang S, Liu C, Ye D, Qi J, Xing Y, Wang J, Fan X, Li X

- Source: PubMed

- PMID: 40339869

- Link: https://pubmed.ncbi.nlm.nih.gov/40339869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning Prediction of 90-Day Mortality in HBV-Related ACLF Using Olink-Derived Inflammatory Protein Signatures.

- Authors: Zhang Y, Sun L, Liu H, Shi K, Lu Y, Feng Y, Wang X

- Source: PubMed

- PMID: 41255250

- Link: https://pubmed.ncbi.nlm.nih.gov/41255250/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning driven prediction of drug efficacy in lung cancer: based on protein biomarkers and clinical features.

- Authors: Li J, Chen A, Liu Z, Wei S, Zhang J, Chen J, Shi C

- Source: PubMed

- PMID: 40355026

- Link: https://pubmed.ncbi.nlm.nih.gov/40355026/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Sequence-based virtual screening using transformers.

- Authors: Zhang S, Huo D, Horne RI, Qi Y, Pujalte Ojeda S, Yan A, Vendruscolo M

- Source: PubMed

- PMID: 40721411

- Link: https://pubmed.ncbi.nlm.nih.gov/40721411/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Computational modelling of olfactory receptors.

- Authors: Odoemelam CS, Steuber V, Schmuker M

- Source: PubMed

- PMID: 40441539

- Link: https://pubmed.ncbi.nlm.nih.gov/40441539/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction

- Authors: Saro Passaro, Gabriele Corso, Jeremy Wohlwend, Mateo Reveiz, Stephan Thaler, Vignesh Ram Somnath, Noah Getz, Tally Portnoi

- Source: OpenAlex

- DOI: 10.1101/2025.06.14.659707

- Link: https://doi.org/10.1101/2025.06.14.659707

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2024

AI-Driven Deep Learning Techniques in Protein Structure Prediction.

- Authors: Chen L, Li Q, Nasif KFA, Xie Y, Deng B, Niu S, Pouriyeh S, Dai Z

- Source: PubMed

- PMID: 39125995

- Link: https://pubmed.ncbi.nlm.nih.gov/39125995/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine learning-based reproducible prediction of type 2 diabetes subtypes.

- Authors: Tanabe H, Sato M, Miyake A, Shimajiri Y, Ojima T, Narita A, Saito H, Tanaka K

- Source: PubMed

- PMID: 39168869

- Link: https://pubmed.ncbi.nlm.nih.gov/39168869/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AKT and EZH2 inhibitors kill TNBCs by hijacking mechanisms of involution.

- Authors: Schade AE, Perurena N, Yang Y, Rodriguez CL, Krishnan A, Gardner A, Loi P, Xu Y

- Source: PubMed

- PMID: 39385030

- Link: https://pubmed.ncbi.nlm.nih.gov/39385030/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning Methods in Protein-Protein Docking.

- Authors: Michalik I, Kuder KJ

- Source: PubMed

- PMID: 38987466

- Link: https://pubmed.ncbi.nlm.nih.gov/38987466/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Virtual patient analysis identifies strategies to improve the performance of predictive biomarkers for PD-1 blockade.

- Authors: Arulraj T, Wang H, Deshpande A, Varadhan R, Emens LA, Jaffee EM, Fertig EJ, Santa-Maria CA

- Source: PubMed

- PMID: 39467131

- Link: https://pubmed.ncbi.nlm.nih.gov/39467131/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Multi-Omics Integration With Machine Learning Identified Early Diabetic Retinopathy, Diabetic Macula Edema and Anti-VEGF Treatment Response.

- Authors: Pang Y, Luo C, Zhang Q, Zhang X, Liao N, Ji Y, Mi L, Gan Y

- Source: PubMed

- PMID: 39671223

- Link: https://pubmed.ncbi.nlm.nih.gov/39671223/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Accurate structure prediction of biomolecular interactions with AlphaFold 3

- Authors: Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore

- Source: OpenAlex

- DOI: 10.1038/s41586-024-07487-w

- Link: https://doi.org/10.1038/s41586-024-07487-w

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Practical guide to SHAP analysis: Explaining supervised machine learning model predictions in drug development

- Authors: Ana Victoria Ponce Bobadilla, Vanessa Schmitt, Corinna S. Maier, Sven Mensing, Sven Stodtmann

- Source: OpenAlex

- DOI: 10.1111/cts.70056

- Link: https://doi.org/10.1111/cts.70056

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2023

Hierarchical graph learning for protein-protein interaction.

- Authors: Gao Z, Jiang C, Zhang J, Jiang X, Li L, Zhao P, Yang H, Huang Y

- Source: PubMed

- PMID: 36841846

- Link: https://pubmed.ncbi.nlm.nih.gov/36841846/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Novel machine learning approaches revolutionize protein knowledge.

- Authors: Bordin N, Dallago C, Heinzinger M, Kim S, Littmann M, Rauer C, Steinegger M, Rost B

- Source: PubMed

- PMID: 36504138

- Link: https://pubmed.ncbi.nlm.nih.gov/36504138/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences

- Authors: Mihály Váradi, Damian Bertoni, Paulyna Magaña, Urmila Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Sreenath Nair

- Source: OpenAlex

- DOI: 10.1093/nar/gkad1011

- Link: https://doi.org/10.1093/nar/gkad1011

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Leakage and the reproducibility crisis in machine-learning-based science

- Authors: Sayash Kapoor, Arvind Narayanan

- Source: OpenAlex

- DOI: 10.1016/j.patter.2023.100804

- Link: https://doi.org/10.1016/j.patter.2023.100804

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Small data machine learning in materials science

- Authors: Pengcheng Xu, Xiaobo Ji, Minjie Li, Wencong Lu

- Source: OpenAlex

- DOI: 10.1038/s41524-023-01000-z

- Link: https://doi.org/10.1038/s41524-023-01000-z

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Generative AI

- Authors: Stefan Feuerriegel, Jochen Hartmann, Christian Janiesch, Patrick Zschech

- Source: OpenAlex

- DOI: 10.1007/s12599-023-00834-7

- Link: https://doi.org/10.1007/s12599-023-00834-7

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2022

Machine Learning Approaches for Metalloproteins.

- Authors: Yu Y, Wang R, Teo RD

- Source: PubMed

- PMID: 35209064

- Link: https://pubmed.ncbi.nlm.nih.gov/35209064/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

AI-Based Protein Structure Prediction in Drug Discovery: Impacts and Challenges.

- Authors: Schauperl M, Denny RA

- Source: PubMed

- PMID: 35727311

- Link: https://pubmed.ncbi.nlm.nih.gov/35727311/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Robust deep learning–based protein sequence design using ProteinMPNN

- Authors: Justas Dauparas, Ivan Anishchenko, Nathaniel R. Bennett, Hua Bai, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Alexis Courbet

- Source: OpenAlex

- DOI: 10.1126/science.add2187

- Link: https://doi.org/10.1126/science.add2187

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Ensemble deep learning: A review

- Authors: M. A. Ganaie, Minghui Hu, A. K. Malik, M. Tanveer, Ponnuthurai Nagaratnam Suganthan

- Source: OpenAlex

- DOI: 10.1016/j.engappai.2022.105151

- Link: https://doi.org/10.1016/j.engappai.2022.105151

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

2021

Artificial intelligence guided discovery of a barrier-protective therapy in inflammatory bowel disease.

- Authors: Sahoo D, Swanson L, Sayed IM, Katkar GD, Ibeawuchi SR, Mittal Y, Pranadinata RF, Tindle C

- Source: PubMed

- PMID: 34253728

- Link: https://pubmed.ncbi.nlm.nih.gov/34253728/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Predictive modeling of estrogen receptor agonism, antagonism, and binding activities using machine- and deep-learning approaches.

- Authors: Ciallella HL, Russo DP, Aleksunes LM, Grimm FA, Zhu H

- Source: PubMed

- PMID: 32778734

- Link: https://pubmed.ncbi.nlm.nih.gov/32778734/

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Highly accurate protein structure prediction with AlphaFold

- Authors: John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates

- Source: OpenAlex

- DOI: 10.1038/s41586-021-03819-2

- Link: https://doi.org/10.1038/s41586-021-03819-2

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

A guide to machine learning for biologists

- Authors: Joe G. Greener, Shaun M. Kandathil, Lewis Moffat, David T. Jones

- Source: OpenAlex

- DOI: 10.1038/s41580-021-00407-0

- Link: https://doi.org/10.1038/s41580-021-00407-0

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Review of deep learning: concepts, CNN architectures, challenges, applications, future directions

- Authors: Laith Alzubaidi, Jinglan Zhang, Amjad J. Humaidi, Ayad Q. Al-Dujaili, Ye Duan, Omran Al-Shamma, José Santamaría, Mohammed A. Fadhel

- Source: OpenAlex

- DOI: 10.1186/s40537-021-00444-8

- Link: https://doi.org/10.1186/s40537-021-00444-8

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Structure-based protein function prediction using graph convolutional networks

- Authors: Vladimir Gligorijevi■, P. Douglas Renfrew, Tomasz Ko■ció■ek, Julia Koehler Leman, Daniel Berenberg, Tommi Vatanen, Chris Chandler, Bryn C. Taylor

- Source: OpenAlex

- DOI: 10.1038/s41467-021-23303-9

- Link: https://doi.org/10.1038/s41467-021-23303-9

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Swarm Learning for decentralized and confidential clinical machine learning

- Authors: Stefanie Warnat■Herresthal, Hartmut Schultze, Krishnaprasad Lingadahalli Shastry, Sathyanarayanan Manamohan, Saikat Mukherjee, Vishesh Garg, Ravi Sarveswara, Kristian Händler

- Source: OpenAlex

- DOI: 10.1038/s41586-021-03583-3

- Link: https://doi.org/10.1038/s41586-021-03583-3

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Using machine learning approaches for multi-omics data analysis: A review

- Authors: Parminder Singh Reel, Smarti Reel, Ewan R. Pearson, Emanuele Trucco, Emily Jefferson

- Source: OpenAlex

- DOI: 10.1016/j.biotechadv.2021.107739

- Link: https://doi.org/10.1016/j.biotechadv.2021.107739

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Machine Learning in Agriculture: A Comprehensive Updated Review

- Authors: Lefteris Benos, Aristotelis C. Tagarakis, Georgios Dolias, Remigio Berruto, Dimitrios Kater■s, Dionysis Bochtis

- Source: OpenAlex

- DOI: 10.3390/s21113758

- Link: https://doi.org/10.3390/s21113758

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Informed Machine Learning - A Taxonomy and Survey of Integrating Prior Knowledge into Learning Systems

- Authors: Laura von Rueden, Sebastian Mayer, Katharina Beckh, Bogdan Georgiev, Sven Giesselbach, Raoul Heese, Birgit Kirsch, Micha■ Walczak

- Source: OpenAlex

- DOI: 10.1109/tkde.2021.3079836

- Link: https://doi.org/10.1109/tkde.2021.3079836

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

The Role of AI, Machine Learning, and Big Data in Digital Twinning: A Systematic Literature Review, Challenges, and Opportunities

- Authors: M. Mazhar Rathore, Syed Attique Shah, Dhirendra Shukla, Elmahdi Bentafat, Spiridon Bakiras

- Source: OpenAlex

- DOI: 10.1109/access.2021.3060863

- Link: https://doi.org/10.1109/access.2021.3060863

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

A review on extreme learning machine

- Authors: Jian Wang, Siyuan Lu, Shuihua Wang■, Yudong Zhang

- Source: OpenAlex

- DOI: 10.1007/s11042-021-11007-7

- Link: https://doi.org/10.1007/s11042-021-11007-7

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Prediction of Chronic Kidney Disease - A Machine Learning Perspective

- Authors: Pankaj Chittora, Sandeep Chaurasia, Pr■sun Chakrabarti, Gaurav Kumawat, Tulika Chakrabarti, Zbigniew Leonowicz, Micha■ Jasi■ski, ■ukasz Jasi■ski

- Source: OpenAlex

- DOI: 10.1109/access.2021.3053763

- Link: https://doi.org/10.1109/access.2021.3053763

- Key finding / methods / datasets / limitations: **Extract from original paper after reading.**

Recommended Next Steps

1. Open the attached individual PDFs (or follow the DOI/URL links).

2. For each relevant paper, extract: main method, datasets used, evaluation metrics, key quantitative results, and stated limitations.

3. Record contradictions or gaps across papers.

4. Only after verification, move evidence into a formal manuscript.

Scientific Rule

AI is an assistant, not a source. Never submit an AI-generated or placeholder claim without checking the original paper.

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